ECE 332 Fall 2025 — Exam 1 Answer Key

[CS p.1: SECTION] = page 1 of cheat sheet; [CS p.2: SECTION] = page 2.

Two corrected errors from prior answer keys:
(1) Q1(c): The exam writes "$\omega = 2\,\text{MHz}$" but MHz is a *frequency* unit. Must convert first: $\omega = 2\pi f = 2\pi \times 2 \times 10^6 = 4\pi \times 10^6$ rad/s. Both prior keys skipped this step and used 2×10^6 directly.
(2) Q6: The exam gives $f = \frac{2}{\pi}$ GHz, not $\frac{2}{5}$ GHz. $\omega = 2\pi f = 2\pi \cdot \frac{2}{\pi} \times 10^9 = 4 \times 10^9$ rad/s.

Question 1 — Triangular Loop (20 pts)

Setup. Equilateral loop in y-z plane, N turns, side a. $\mathbf{B}(t) = B_0(-2\hat{x} + 3\hat{y})\sin(\omega t)$.

(a) Magnetic flux through one turn. *(6 pts)*

[CS p.1: SURFACE ELEMENT $d\mathbf{s}$]: loop in y-z plane $\Rightarrow$ $d\mathbf{s} = \hat{x}\,da$.
[CS p.1: DOT PRODUCTS]: $\hat{x}\cdot\hat{x} = 1$, $\hat{y}\cdot\hat{x} = 0$ $\Rightarrow$ only the $-2\hat{x}$ component contributes.
[CS p.1: ROTATING LOOP area table]: equilateral triangle with side a: $A = \frac{\sqrt{3}}{4}a^2$.

$$\Phi = \int_S \mathbf{B}\cdot d\mathbf{s} = B_0(-2\underbrace{\hat{x}\cdot\hat{x}}_{1} + 3\underbrace{\hat{y}\cdot\hat{x}}_{0})\sin(\omega t)\cdot A$$

$$\boxed{\Phi = -\tfrac{\sqrt{3}}{2}\,B_0\,a^2\sin(\omega t)\text{ Wb}}$$

(b) EMF induced in the loop. *(6 pts)*

[CS p.1: EMF & FARADAY'S LAW]: $V_{\text{emf}} = -N\dfrac{d\Phi}{dt}$.
[CS p.1: $\partial B/\partial t$ TABLE]: $\dfrac{d}{dt}[\sin(\omega t)] = \omega\cos(\omega t)$.

$$V_{\text{emf}} = -N\,\frac{d}{dt}\left[-\tfrac{\sqrt{3}}{2}B_0 a^2\sin(\omega t)\right]$$

$$\boxed{V_{\text{emf}} = \frac{\sqrt{3}}{2}\,N B_0\,\omega\,a^2\cos(\omega t)\text{ V}}$$

(c) Current through $1\,\text{k}\Omega$ at given values. *(8 pts)*

Unit conversion required. The exam states "$\omega = 2\,\text{MHz}$", but MHz is a frequency unit (cycles/s). [CS p.1: CIRCUIT & WAVE strip]: $\omega = 2\pi f$, so

$$\omega = 2\pi \times 2 \times 10^6 = 4\pi \times 10^6\text{ rad/s}$$

[CS p.1: SI PREFIXES]: MHz $= 10^6$.

Given: $N = 10$, $a = 0.08\,\text{m}$, $B_0 = 0.02\,\text{T}$, $\omega = 4\pi \times 10^6$ rad/s, $\omega t = \pi/4$, $R = 1\,\text{k}\Omega$.

Step 1. Combine trig factors.
[CS p.1: UNIT CIRCLE]: $\cos(\pi/4) = \frac{\sqrt{2}}{2}$.

$$\frac{\sqrt{3}}{2}\cdot\frac{\sqrt{2}}{2} = \frac{\sqrt{6}}{4}$$

Step 2. Compute $|V_{\text{emf}}|$ numerically.

$$|V_{\text{emf}}| = \frac{\sqrt{6}}{4} \cdot N \cdot B_0 \cdot \omega \cdot a^2$$
$$= \frac{\sqrt{6}}{4} \cdot 10 \cdot 0.02 \cdot (4\pi \times 10^6) \cdot (0.08)^2$$

Bracket step by step:

$$(0.08)^2 = 6.4 \times 10^{-3}, \quad 0.02 \times 6.4 \times 10^{-3} = 1.28 \times 10^{-3}$$

$$1.28 \times 10^{-3} \times 4\pi \times 10^6 = 5.12\pi \times 10^3, \quad 10 \times 5.12\pi \times 10^3 = 5120\pi \times 10^1$$

Wait, step more carefully:

$$10 \times 1.28 \times 10^{-3} \times 4\pi \times 10^6 = 10 \times 5.12\pi \times 10^3 = 51200\pi$$

$$|V_{\text{emf}}| = \frac{\sqrt{6}}{4} \times 51200\pi = 12800\pi\sqrt{6} \approx 12800 \times 3.1416 \times 2.449 \approx 98{,}490 \text{ V}$$

Hmm — let me redo carefully:

$$10 \times 0.02 = 0.2, \quad 0.2 \times (0.08)^2 = 0.2 \times 6.4 \times 10^{-3} = 1.28 \times 10^{-3}$$

$$1.28 \times 10^{-3} \times 4\pi \times 10^6 = 5120\pi \text{ (not } 5120\pi \times 10^1\text{)}$$

$$|V_{\text{emf}}| = \frac{\sqrt{6}}{4} \times 5120\pi = 1280\pi\sqrt{6}$$

$$1280 \times \pi \approx 4021, \quad 4021 \times \sqrt{6} \approx 4021 \times 2.449 \approx 9847 \text{ V}$$

$$\boxed{|V_{\text{emf}}| = 1280\pi\sqrt{6} \approx 9847 \text{ V}}$$

Step 3. Apply Ohm's law [CS p.1: EMF & FARADAY'S LAW equivalent circuit].

$$I = \frac{V_{\text{emf}}}{R} = \frac{1280\pi\sqrt{6}}{1000} = 1.28\pi\sqrt{6}$$

$$\boxed{I = 1.28\pi\sqrt{6} \approx 9.85 \text{ A}}$$

Compare: prior keys got ≈ 1.57 A by omitting the 2π conversion. The factor of ≈ 6.28 difference is exactly 2π.

Question 2 — Moving Wire (12 pts)

Wire along $\hat{y}$, $a = 0.15$ m, $\mathbf{u} = 0.12\hat{x}$ m/s, $\mathbf{B} = (0.1\hat{x} - 0.05\hat{z})$ T.

[CS p.1: EMF & FARADAY'S LAW (Motional EMF)]: $V_{\text{emf}} = \int (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$.

Step 1. Compute $\mathbf{u} \times \mathbf{B}$ [CS p.1: CROSS PRODUCT (determinant)].

$$\mathbf{u} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0.12 & 0 & 0 \\ 0.1 & 0 & -0.05 \end{vmatrix} = \hat{x}(0 \cdot (-0.05) - 0 \cdot 0) - \hat{y}(0.12 \cdot (-0.05) - 0 \cdot 0.1) + \hat{z}(0.12 \cdot 0 - 0 \cdot 0.1)$$

$$= \hat{x}(0) - \hat{y}(-0.006) + \hat{z}(0) = 0.006\hat{y}\ \text{V/m}$$

Step 2. Integrate along wire ($d\mathbf{l} = \hat{y}\,dy$).

$$V_{\text{emf}} = \int_0^{0.15} 0.006\,dy = 0.006 \times 0.15$$

$$\boxed{V_{\text{emf}} = 9 \times 10^{-4}\ \text{V} = 0.9\ \text{mV}}$$

Only the $-0.05\hat{z}$ component of $\mathbf{B}$ contributes; the $\hat{x} \times \hat{x} = 0$ term drops out. Common error: forgetting to convert cm/s $\to$ m/s (factor-of-100 mistake).

Question 3 — Conduction & Displacement Current (12 pts)

$I_c = 2\sin(\omega t)\,\mu$A, cross-section A, $\sigma = 10^{-4}$ S/m, $\varepsilon_r = 26\pi$.

(a) Electric field E. *(6 pts)*

[CS p.1: DISPLACEMENT & CONDUCTION CURRENT]: $J_c = \sigma E$, $I_c = J_c \cdot A$.

$$J_c = \frac{I_c}{A}, \qquad E = \frac{J_c}{\sigma} = \frac{I_c}{\sigma A} = \frac{2 \times 10^{-6}\sin(\omega t)}{10^{-4} \cdot A}$$

$$\boxed{E = \frac{0.02}{A}\sin(\omega t)\ \text{V/m}}$$

Area A remains in the answer — the problem gives no numeric value for it.

(b) Displacement current I_d at $\omega = 10^9$ rad/s, $\omega t = \pi/3$. *(6 pts)*

[CS p.1: DISPLACEMENT & CONDUCTION CURRENT]: $J_d = \varepsilon\,\partial E/\partial t$, $I_d = J_d \cdot A$.
[CS p.1: $\partial B/\partial t$ TABLE]: $\partial[\sin(\omega t)]/\partial t = \omega\cos(\omega t)$.
[CS p.1: UNIT CIRCLE]: $\cos(\pi/3) = \frac{1}{2}$.

Step 1. Compute $\varepsilon = \varepsilon_r\varepsilon_0$.

$$\varepsilon = 26\pi \cdot \frac{1}{36\pi} \times 10^{-9} = \frac{26}{36} \times 10^{-9} = \frac{13}{18} \times 10^{-9}\ \text{F/m}$$

Step 2. $I_d = \varepsilon A \cdot \dfrac{\partial E}{\partial t} = \varepsilon A \cdot \dfrac{0.02}{A} \cdot \omega\cos(\omega t)$ (the A cancels).

$$I_d = \frac{13}{18} \times 10^{-9} \times 0.02 \times 10^9 \times \frac{1}{2} = \frac{13}{18} \times 0.02 \times \frac{1}{2} = \frac{13}{1800}$$

$$\boxed{I_d = \frac{13}{1800} \approx 7.22\ \text{mA}}$$

Question 4 — Charge Relaxation (10 pts)

$\sigma = 10^{-4}$ S/m, $\varepsilon_r = 26\pi$.

(a) Relaxation time constant. *(5 pts)*

[CS p.1: CHARGE RELAXATION]: $\tau_r = \varepsilon_r \varepsilon_0 / \sigma$.

$$\tau_r = \frac{26\pi \cdot \frac{1}{36\pi} \times 10^{-9}}{10^{-4}} = \frac{\frac{13}{18} \times 10^{-9}}{10^{-4}} = \frac{13}{18} \times 10^{-5}\ \text{s}$$

$$\boxed{\tau_r = \frac{13}{18} \times 10^{-5} \approx 7.22\ \mu\text{s}}$$

(b) Time to decay to e^{-3} of initial value. *(5 pts)*

[CS p.1: CHARGE RELAXATION]: $\rho_v(t) = \rho_{v0}\, e^{-t/\tau_r}$.

$$e^{-t/\tau_r} = e^{-3} \ \Rightarrow\ \frac{t}{\tau_r} = 3 \ \Rightarrow\ t = 3\tau_r$$

$$\boxed{t = 3 \times 7.22\ \mu\text{s} \approx 21.7\ \mu\text{s}}$$

Question 5 — Lossless EM Wave (30 pts)

$\mathbf{E}(z,t) = 4\cos\bigl(6\pi \times 10^8\, t - kz - \pi/6\bigr)\hat{y}$ V/m. $\varepsilon_r = 6$, $\mu_r = 1$.

(a) Find ω, k, λ, η. *(10 pts)*

[CS p.2: PLANE WAVE PARAMETERS (LOSSLESS)]: read ω from the cosine argument; chain $\omega \to u_p \to k \to \lambda \to \eta$.

$$\omega = 6\pi \times 10^8\ \text{rad/s}$$

$$k = \frac{\omega\sqrt{\varepsilon_r}}{c} = \frac{6\pi \times 10^8 \cdot \sqrt{6}}{3 \times 10^8} = 2\pi\sqrt{6} \qquad \boxed{k = 2\pi\sqrt{6} \approx 15.4\ \text{rad/m}}$$

$$\lambda = \frac{2\pi}{k} = \frac{2\pi}{2\pi\sqrt{6}} = \frac{1}{\sqrt{6}} \qquad \boxed{\lambda = \frac{1}{\sqrt{6}} \approx 0.408\ \text{m}}$$

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{6}} = 20\pi\sqrt{6} \qquad \boxed{\eta = 20\pi\sqrt{6} \approx 154\ \Omega}$$

(b) Find $\tilde{\mathbf{E}}(z)$. *(5 pts)*

[CS p.1: PHASOR METHOD]: $\mathbf{E} = \text{Re}\{\tilde{\mathbf{E}}\, e^{j\omega t}\}$; strip $e^{j\omega t}$, keep the rest.
[CS p.2: PHASOR ↔ TIME DOMAIN]: $Ae^{j\phi}e^{-j\beta z} \leftrightarrow A\cos(\omega t - \beta z + \phi)$.

$$\boxed{\tilde{\mathbf{E}}(z) = 4\, e^{-j(kz + \pi/6)}\ \hat{y}\ \text{V/m}}$$

(c) Find $\tilde{\mathbf{H}}(z)$. *(5 pts)*

[CS p.2: DIRECTION OF PROPAGATION]: argument contains $-kz \Rightarrow \hat{k} = +\hat{z}$.
[CS p.2: $\hat{k} \times \hat{E}$ — FIND $\hat{H}$ DIRECTION]: $\hat{k} = +\hat{z}$, $\hat{E} = \hat{y} \Rightarrow \hat{z} \times \hat{y} = -\hat{x}$.
[CS p.2: LOSSLESS vs. LOSSY — E & H]: $\tilde{\mathbf{H}} = \frac{1}{\eta}(\hat{k} \times \tilde{\mathbf{E}})$.

$$\tilde{\mathbf{H}} = \frac{1}{\eta}\,\hat{z} \times \left(4e^{-j(kz+\pi/6)}\hat{y}\right) = \frac{4}{\eta}\, e^{-j(kz+\pi/6)}\,(\hat{z} \times \hat{y})$$

With $\eta = 20\pi\sqrt{6}$: $\frac{4}{\eta} = \frac{4}{20\pi\sqrt{6}} = \frac{1}{5\pi\sqrt{6}}$.

$$\boxed{\tilde{\mathbf{H}}(z) = -\frac{1}{5\pi\sqrt{6}}\, e^{-j(kz+\pi/6)}\,\hat{x}\text{ A/m} \approx -0.026\, e^{-j(kz+\pi/6)}\,\hat{x}\text{ A/m}}$$

(d) Find $\mathbf{H}(z, t)$. *(5 pts)*

[CS p.2: PHASOR ↔ TIME DOMAIN]: $\text{Re}\{Ae^{j\phi}e^{j\omega t}e^{-j\beta z}\} = A\cos(\omega t - \beta z + \phi)$.

$$\boxed{\mathbf{H}(z,t) = -\frac{1}{5\pi\sqrt{6}}\cos\left(6\pi\times 10^8\, t - 2\pi\sqrt{6}\, z - \frac{\pi}{6}\right)\hat{x}\text{ A/m}}$$

(e) Average power density. *(5 pts)*

[CS p.2: AVERAGE POWER DENSITY (POYNTING) lossless shortcut]: $\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta}\,\hat{k}$.
$|E_0| = 4\text{ V/m}$, $\hat{k} = +\hat{z}$, $\eta = 20\pi\sqrt{6}$:

$$\mathbf{S}_{\text{av}} = \frac{16}{2\times 20\pi\sqrt{6}}\,\hat{z} = \frac{16}{40\pi\sqrt{6}}\,\hat{z} = \frac{2}{5\pi\sqrt{6}}\,\hat{z} = \frac{\sqrt{6}}{15\pi}\,\hat{z}$$

$$\boxed{\mathbf{S}_{\text{av}} = \frac{\sqrt{6}}{15\pi}\,\hat{z} \approx 0.052\,\hat{z}\text{ W/m}^2}$$

Question 6 — EM Wave in Distilled Water (16 pts)

Given: $f = \frac{2}{\pi}$ GHz, $\sigma = 10^{-4}$ S/m, $\varepsilon_r = 26\pi$, $\mu_r = 1$.

Compute ω first [CS p.2: WAVE PARAMS strip]: $\omega = 2\pi f$

$$\omega = 2\pi\times\frac{2}{\pi}\times 10^9 = 4\times 10^9\text{ rad/s}$$

Also useful: $\varepsilon' = \varepsilon_r\varepsilon_0 = 26\pi\cdot\frac{1}{36\pi}\times 10^{-9} = \frac{13}{18}\times 10^{-9}$ F/m

(a) Find $\varepsilon''/\varepsilon'$. *(6 pts)*

[CS p.2: LOSSY MEDIA — CLASSIFICATION]: $\varepsilon''/\varepsilon' = \sigma/(\omega\varepsilon_r\varepsilon_0)$.

$$\frac{\varepsilon''}{\varepsilon'} = \frac{10^{-4}}{4\times 10^9\times\frac{13}{18}\times 10^{-9}} = \frac{10^{-4}}{\frac{52}{18}} = \frac{10^{-4}\times 18}{52} = \frac{18}{52}\times 10^{-4} = \frac{9}{26}\times 10^{-4}$$

$$\boxed{\frac{\varepsilon''}{\varepsilon'} = \frac{9}{26}\times 10^{-4} \approx 3.46\times 10^{-5}}$$

(b) What type of medium? *(5 pts)*

[CS p.2: LOSSY MEDIA — CLASSIFICATION table]: threshold for low-loss is $\varepsilon''/\varepsilon' < 0.01$.

$$\frac{\varepsilon''}{\varepsilon'} \approx 3.46\times 10^{-5} \ll 0.01$$

$$\boxed{\text{Low-loss dielectric.}}$$

Justification: the ratio is far below the 10^{-2} threshold.

(c) Skin depth. *(5 pts)*

[CS p.2: LOW-LOSS MEDIUM]: $\alpha \approx \frac{\sigma}{2}\sqrt{\mu/\varepsilon'} = \frac{\sigma\eta}{2}$.
[CS p.2: SKIN DEPTH]: $\delta_s = 1/\alpha$.

Step 1. Find η [CS p.2: WAVE PARAMS strip].

$$\eta = \frac{\eta_0}{\sqrt{\varepsilon_r}} = \frac{120\pi}{\sqrt{26\pi}} \approx \frac{376.99}{9.04} \approx 41.7\ \Omega$$

Step 2. Attenuation constant (low-loss approximation).

$$\alpha \approx \frac{\sigma\eta}{2} = \frac{10^{-4} \times 41.7}{2} \approx 2.09 \times 10^{-3}\ \text{Np/m}$$

Step 3. Skin depth.

$$\delta_s = \frac{1}{\alpha} = \frac{1}{2.09 \times 10^{-3}}$$

$$\boxed{\delta_s \approx 479\ \text{m}}$$

≈ 480 m confirms this is essentially lossless at $\frac{2}{\pi}$ GHz with $\sigma = 10^{-4}$ S/m.

Score (if perfect): Q1: 20/20 Q2: 12/12 Q3: 12/12 Q4: 10/10 Q5: 30/30 Q6: 16/16 = **100/100**